

Lecture 6: Real numbers

How to manage different & relations between them?

Situation: $n: \mathbb{N}$, $x: \mathbb{Q}$, $f: \mathbb{R} \rightarrow \mathbb{R}$.

Want to evaluate f at $\underline{x + \frac{1}{n}}$.

Difficulties:

1) What is $\frac{1}{n}$? (if $n \geq 1$)
 $\frac{1}{n} = 0$.
in lean w/o extra info
A: Tell lean: $\frac{1}{n}: \mathbb{Q}$.

2) But f expects a Real number!

We have to convert $x + \frac{1}{n}$ to real number!

Two ways of doing so! i) $(x + \frac{1}{n}): \mathbb{R}$

ii) $(x: \mathbb{R}) + (\frac{1}{n}: \mathbb{R}): \mathbb{R}$.

We need to be sure that these two ways are the same!

Naive solution: In lean we write $f((x: \mathbb{R}) + ((\frac{1}{n}: \mathbb{Q}): \mathbb{R}))$

Smart solution: Is a type system that allows us to automatically change the type of variable based on context.
→ Coercion!

Upshot: We can simply write $f(x + \frac{1}{n})$ and
learn figure out what to do!

How to define our own types?

We can build new types from the types we know

- $\mathbb{N}, \mathbb{Q}, \mathbb{R}, \dots$
- Basic operations give any two types, X, Y
 $X \sqcup Y, X \times Y, X \rightarrow Y$.
- Combine different types into new structures

→ structured types.

Ex We want to define a type whose terms

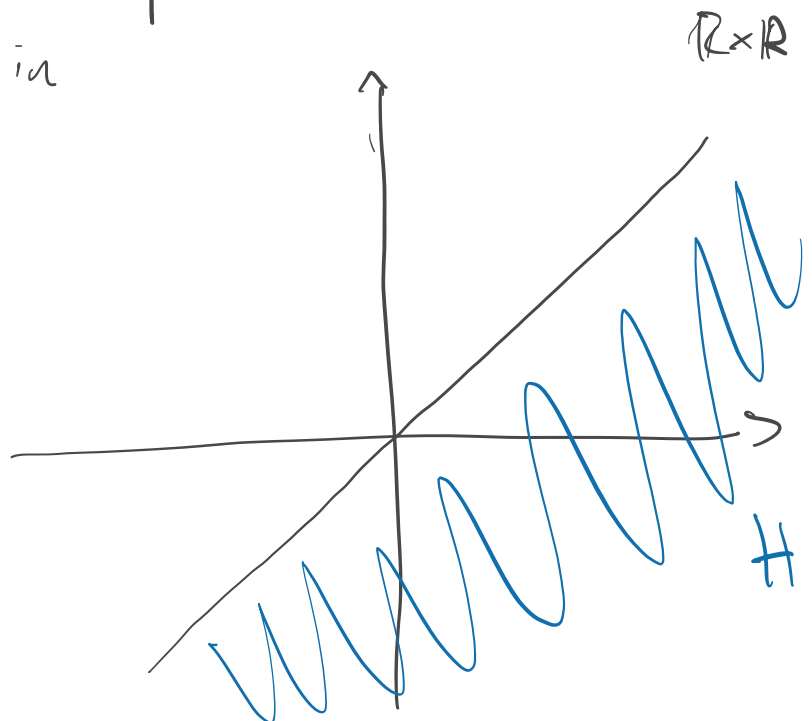
correspond to point in
in $\mathbb{R} \times \mathbb{R}$:

structure H where

$x: \mathbb{R}$

$y: \mathbb{R}$

$x \leq y: y \leq x$



Observation: There is a canonical map

$$H \longrightarrow \mathbb{R} \times \mathbb{R}$$

$$(x, y, x - 4t - y) \longmapsto (x, y).$$

$\Rightarrow$ We want to tell him that it can automatically convert $(p: H)$ to a point $(p: \mathbb{R}^2)!$

Careful: It doesn't make sense to automatically convert $(p: H)$ to a real number!!

The real numbers:

Ex. Q: Is there a number x s.t. $x^2 = 2$?

1) $x: \mathbb{Z}$. x^2 preserves integ.

But $1^2 = 1 \wedge 2^2 = 4$. so no such x !

2) A rational number $x = \frac{n}{m}$ s.t. $x^2 = 2$?

A: No! Proof by Contradiction!

Observation: $a \in \mathbb{N}$. Then $\text{PEXP}(a^2)(p)$ is even!
 $p \in \mathbb{N}$ prime $= \text{PEXP}(a)(p) + \text{PEXP}(a)(p)$

Assume: $x = \frac{n}{m}$ s.t. $x^2 = 2$.

Then $n^2 = 2m^2$. furthermore. pick p prime number

$$\text{PEXP}(n^2)(p) = \text{PEXP}(m^2 \cdot 2)(p) = \text{PEXP}(m^2)(p) + \text{PEXP}(2)(p)$$

For $p=2$. $\downarrow$ even $\text{PEXP}(m^2)(2) + 2$.
 $\downarrow$ even

3) Should there be such a number x ?

Ans: Yes, if we imagine the numbers to form a continuous line!

How to find such an x (assuming the existence of a "continuous" set of numbers containing $\mathbb{Q}$)

Consider the following sets I_n for $n \in \mathbb{N}$.

- $I_0 = [1, 2]$.

- I_1 : $\frac{1+2}{2} = \frac{3}{2}$, check $(\frac{3}{2})^2 = \frac{9}{4} > 2$.

we define $I_1 = [1, \frac{3}{2}]$.

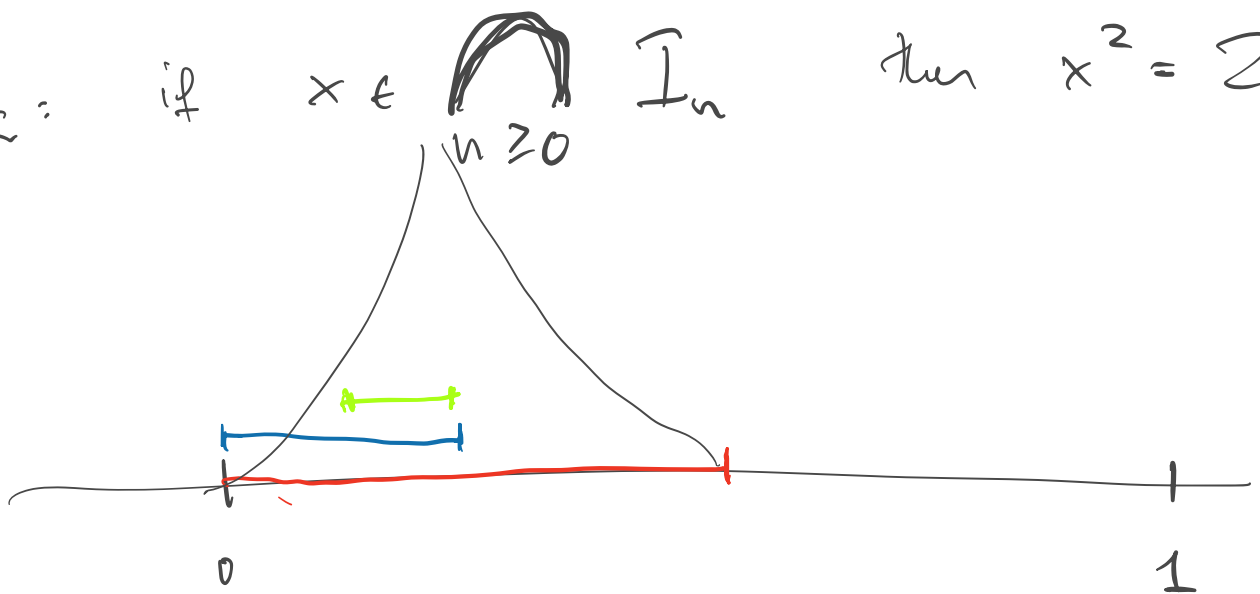
- by induction: assume we have $I_n = [a_n, b_n]$.
 check if $\left(\frac{a_n + b_n}{2}\right)^2 > 2$ or $\left(\frac{a_n + b_n}{2}\right)^2 < 2$.

$$I_{n+1} = \left[a_n, \frac{a_n + b_n}{2}\right]$$

$$I_{n+1} = \left[\frac{a_n + b_n}{2}, b_n\right]$$

Finally $I_{n+1} \subset I_n$

Claim: if $x \in \bigcap_{n \geq 0} I_n$ then $x^2 = 2$.



Formal definition of the real numbers

Def 1) A sequence of rational (real) is a map
 $x: \mathbb{N} \rightarrow \mathbb{Q}$ we usually x_n instead of $x(n)$

2) A sequence x of rat. numbers converges to $a \in \mathbb{Q}$
 if $\forall \varepsilon > 0 \exists N: \mathbb{N}$ s.t. $\forall n \geq N: |x_n - a| < \varepsilon$.

we usually write $\lim x_n = a$

$n \rightarrow \infty$
 as $x_n \rightarrow a$ as $n \rightarrow \infty$.
 and a is called the limit of x .

3) A sequence $x: \mathbb{N} \rightarrow \mathbb{Q}$ is called Cauchy
 if

$$\forall \epsilon > 0 \exists N: \mathbb{N} \forall n, m \geq N: |x_n - x_m| < \epsilon.$$

Ex. In the example above. Pick $x_n \in I_n \subset \mathbb{Q}$.
 defines a sequence. $x: \mathbb{N} \rightarrow \mathbb{Q}$
 which is Cauchy!

(because $\text{diam}|I_n| \rightarrow 0$ as $n \rightarrow \infty$)
 and $I_{n+1} \subset I_n$

Observation: x_n has no limit point in $\mathbb{Q}$!

Defn. The real numbers $\mathbb{R}$ are defined as the
Cauchy completion of $\mathbb{Q}$, that means
 $\mathbb{Q} \subset \mathbb{R}$ and additionally $\mathbb{R}$ contains a

limit point for every Cauchy seq. of rat. numbers.

Construction: $\{x: \text{rat. Cauchy seq.}\} / x \sim y$
 $x \sim y$ if $|x - y|$ tends to zero.

In other words, it is the smallest
"set" containing $\mathbb{Q}$ and at least
one limit point for every Cauchy seq.

Ex. Assume we have candidate for
the real numbers. $\mathbb{R}$

The following holds. For $x \in \mathbb{R}$
 $x = 0 \iff |x| < \frac{1}{n} \quad \forall n \geq 0.$

Claim. Assume we have $x: \mathbb{N} \rightarrow \mathbb{R}$

s.t. $\lim_{n \rightarrow \infty} x_n = a$ then $a = b$,

and $\lim_{n \rightarrow \infty} x_n = b$

Pf. We want to show $a = b$

as $a - b = 0$,

ISTS: $(a - b) < \frac{1}{n}$ for every $n \geq 0$

Fix $n \geq 0$.

By convergence of x_n there exist

for $\epsilon = \frac{1}{2n}$ a N s.t.

for all $m \geq N$: $|x_m - a| < \epsilon$
 $\leq \frac{1}{2n}$

there exists K st.

for all $n \geq K : (x_n - b) < \epsilon$
 $= \frac{1}{2n}.$

Trich. $|a - b| = |a - x_n + x_n - b|$

(for some $n \geq \max(N, K)$)

$$\leq |a - x_n| + |x_n - b|$$

$$< \frac{1}{2n} + \frac{1}{2n}.$$

$$< \frac{1}{n}.$$
